

12.To Create Pipes, Redirect Streams, Execute Connected Commands, Close Descriptors, Synchronize Processes, Test Data Flow.

To Support Multiple Pipes, Manage Descriptor Arrays, Launch Multiple Processes, Coordinate Execution, Cleanup Resources, Test Long Pipelines.

```
#include <stdio.h>
#include <unistd.h>
#include <sys/wait.h>
#include <stdlib.h>

int main() {
    int fd[2];
    pid_t p1, p2;

    if (pipe(fd) == -1) {
        perror("pipe");
        return 1;
    }

    p1 = fork();

    if (p1 == 0) {
        close(fd[0]);

        dup2(fd[1], STDOUT_FILENO);
        close(fd[1]);

        execlp("ls", "ls", NULL);

        perror("execlp");
        exit(1);
    }

    p2 = fork();

    if (p2 == 0) {
        close(fd[1]);

        dup2(fd[0], STDIN_FILENO);
        close(fd[0]);

        execlp("wc", "wc", "-l", NULL);
```

```
    execlp("wc", "wc", "-l", NULL);

    perror("execlp");
    exit(1);
}

close(fd[0]);
close(fd[1]);

waitpid(p1, NULL, 0);
waitpid(p2, NULL, 0);

return 0;
}
```

```
bindu@Bindusree: ~  
bindu@Bindusree:~$ nano skill12.c  
bindu@Bindusree:~$ gcc skill12.c -o skill12  
bindu@Bindusree:~$ ./skill12  
Simple Command History  
Type 'history' to display commands  
Type 'exit' to quit  
  
shell> 9  
shell> |
```